Dimension-Sensitive Reaction Bounds and Causal Heat Certificates for a Two-Scale Ricci-Flow Construction

Dr. Muralidhara Subbarao
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Stony Brook University
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Abstract

We analyze the local analytic and reaction interfaces in the supplied manuscript *A closed Ricci flow with bounded scalar curvature and finite-time curvature blowup*. Keeping the multiplicity-weighted curvature matrix and its separated dominant block improves its reaction upper bound from $2(q-1)+O(\sqrt{q})$ to $2(q-1)+O(1)$, under the same source derivative estimates. An exact two-dimensional envelope and a sharp algebraic example quantify the gain. We carry it through the actual cap barrier, obtaining a conditional spatial Ricci exponent $e=\kappa/(q-1)+d_q/k$ with one dimension-independent κ. A separate causal heat-inverse certificate records coefficient, forcing and truncation errors in the correct parabolic norms, without differentiating the coefficients. Its maximal-regularity premise is stronger than the private corpus's energy heat theorem and is stated explicitly. Construction, point-picking, cap control and global singularity existence remain imported inputs, not consequences of finite matrix tests. Contribution assessment: C3/5, a source-specific constant-order reaction loss and its quantitative barrier consequences are proved, with explicit causal error control and no independent certification of the global Ricci-flow counterexample.

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1 The geometric model and the source boundary

The source [1] asserts a smooth Ricci flow on a closed manifold $\mathbb{S}^2 \times \mathbb{S}^{q+1}$, in a sufficiently large fixed dimension $q+3$, with bounded scalar curvature and full-curvature blowup at a finite maximal time. This report does not independently establish that assertion. It studies two precise steps of the argument after their geometric hypotheses have been supplied.

Off the two pole orbits the metric is

$$g = \mathrm{d}s^2 + \phi(s,t)^2 g_{\mathbb{S}^2} + r(s,t)^2 g_{\mathbb{S}^q}, \qquad \partial_t g = -2\,\mathrm{Ric}\,.$$

The sphere metrics have sectional curvature one; s is physical arclength at time t, not a fixed material coordinate. At each pole $r = 0, r_s = 1, \phi_s = 0, \phi > 0$, with the usual odd/even smooth polar extensions. Reflection exchanges the poles. On an open hemisphere $r_s > 0$, so r can serve as a coordinate; it cannot do so at the equator. In the pole conditions above, arclength points inward from the indicated pole; one globally oriented arclength has $r_s = -1$ at the other pole. Put

$$\delta = T - t, \quad \theta = \delta^\sigma, \quad d_q = \frac{q+1-\sqrt{(q+1)(q-7)}}{2}, \quad \nu = k - d_q/2, \quad \sigma = k/d_q. \tag{1.1}$$

Here $q \geq 10$, and k is a sufficiently large even integer chosen after q. The cap scale θ and parabolic scale $\sqrt{\delta}$ are distinct. Constants for one chosen flow can depend on both parameters; a constant used to choose q cannot depend on either of them.

With $u = \log\phi$, $z = r_s^2$, the sectional curvatures are

$$\begin{gathered} h = -\phi_{ss}/\phi, \quad h_q = -r_{ss}/r, \quad j = (1-\phi_s^2)/\phi^2, \quad \ell = (1-r_s^2)/r^2, \quad \mu = -\phi_s r_s/(\phi r), \\ h = -z(u_{rr}+u_r^2) - \tfrac{1}{2} z_r u_r, \quad h_q = -z_r/(2r), \quad j = e^{-2u} - z u_r^2, \quad \ell = (1-z)/r^2, \quad \mu = -z u_r/r. \end{gathered} \tag{1.2}$$

The second line is only used where the radius coordinate is valid. For an invariant diagonal tensor with entries $(\alpha, \beta, \beta, \chi^{\times q})$, the physical squared norm is $\alpha^2 + 2\beta^2 + q\chi^2$. Unweighted coordinates would change the operator norm and invalidate the reaction estimate.

1.1 Architecture and retained inputs

All six scientific source sections were freshly read in full, together with the wrapper, figures, bibliography and README. The logical steps are:

1. `02-construction`: Stolarski's singular-flow construction, weighted profiles through two derivatives, inner signs, and the ordered choice of tolerances, cutoffs and starting time. The initial outer exponent and propagated outer exponent are different permitted choices.

2. `03-geometry`: material-time versus fixed-radius equations, global slope bounds, scalar lower bound, strict cone separation, and fixed parabolic-annulus Ricci estimates.

3. `04-cap`: radius and cap curvature bounds by record points, controlled backward neighborhoods, Shi derivative estimates, pole-orbit size and a radial-slope barrier. No convergence to a prescribed cap metric is assumed in this argument.

4. `05-reaction`: a one-dimensional derivative-free interior estimate, moving cylinders canceling the large radial drift, and the physical three-by-three reaction matrix.

5. `06-barriers`: inner Ricci comparison, pole excision for a nonsmooth scalar barrier, exterior comparison across the equator, then finite-time blowup and two-sided curvature rates.

Full text reading is not a proof audit of the external construction or every imported analytic theorem. In particular, none of the finite calculations below constructs a closed Ricci flow.

The source obtains, on $0 < r \le \sqrt{\delta}$,

$$|z-1| \le C_0/q, \qquad r|u_r| \le 2, \qquad r^2 e^{-2u} \le q - 1 + C_0, \tag{1.3}$$

with a numerical C_0 (the source permits $C_0 = 4$). On the smaller region $0 < r \le \sqrt{\delta}/8$, its derivative estimates give

$$|rz_r| \le Z q^{-1/2}, \qquad r^2|u_{rr}| \le U\sqrt{q}, \tag{1.4}$$

where U, Z are independent of q, k, once the universal thresholds are met. We retain these as precise hypotheses, not numerically known constants. The estimates $(r^2+\theta^2)|\operatorname{Rm}| \le C$ and $|\operatorname{Ric}| \le C_{a_1,a_2}\delta^{\nu-1}$ on fixed parabolic annuli are separate geometric inputs with flow-dependent constants.

2 The causal heat step, with its actual strength

On $P_R = (-R, R) \times [-R^2, 0]$, the source treats

$$H_t = aH_{yy} + bH_y + F, \qquad |a-1| \le \epsilon_0, \quad |H| + |b| + |F| \le K.$$

For one fixed $p > 3$, it invokes the classical whole-space parabolic estimate $\|V_{yy}\|_p + \|V_t\|_p \le C_p\|V_t - V_{yy}\|_p$ for a causal, past-zero function. The cited Krylov theorem is an external maximal-regularity input, not a consequence of an energy inequality. Time reversal aligns its backward convention with this forward one [3].

For a cutoff $V = \zeta H$, the exact identity is

$$(\partial_t - \partial_{yy})V = (a-1)V_{yy} + bV_y + \zeta F + (\zeta_t - a\zeta_{yy} - b\zeta_y)H - 2a\zeta_y H_y. \tag{2.1}$$

There are no derivatives of a, b. Absorbing $C_p\epsilon_0 < 1/2$, then using spatial interpolation with a parameter proportional to the cylinder gap, gives the source recurrence

$$E(R_1) \le \tfrac{1}{16}E(R_2) + C(K)(R_2 - R_1)^{-2}.$$

On its nested radii, the errors grow by four per step and the multiplier decays by sixteen. The remaining $E(7/4)$ is finite for each smooth solution; no uniform bound for it is assumed. Summation followed by parabolic embedding with $p > 3$ yields the interior gradient bound. An energy solution merely in $L^2_t H^1_y$, with time derivative in $L^2_t H^{-1}_y$, does not supply that embedding.

2.1 A residual certificate on the causal function space

Let $Q = \mathbb{R} \times (t_0, t_1)$, $t_1 < \infty$, and extend sources by zero before t_0. Norms below use physical $\mathrm{d}y\,\mathrm{d}t$, not normalized probability volume. For $\lambda > 0$, define

$$(T_\lambda f)(y,t) = \int_{t_0}^{t} e^{-\lambda(t-s)}(G_{t-s} * f(\cdot,s))(y)\,\mathrm{d}s, \qquad G_t(y) = (4\pi t)^{-1/2}e^{-y^2/(4t)}.$$

Let A_p be a uniform classical bound for $\|\partial_{yy}T_\lambda\|_{L^p(Q)\to L^p(Q)}$. The constant-coefficient maximal-regularity estimate supplies one independent of the finite time interval and λ. The causal kernel itself also gives

$$\|T_\lambda\|_{p\to p} \le \lambda^{-1}, \qquad \|\partial_y T_\lambda\|_{p\to p} \le \lambda^{-1/2}. \tag{2.2}$$

Indeed $\|G_t\|_1 = 1$, $\|\partial_y G_t\|_1 = (\pi t)^{-1/2}$, and integration with $e^{-\lambda t}$ proves both bounds by Young's inequality.

Theorem 2.1 (Causal coefficient and residual budget)**.** *Let* $1 < p < \infty$, $\|a-1\|_\infty \le \varepsilon < 1$, $\|b\|_\infty \le B$, *and suppose*

$$\kappa = A_p\varepsilon + B/\sqrt{\lambda} < 1. \tag{2.3}$$

For every $g \in L^p(Q)$, *there is a unique past-zero* $v \in W_p^{2,1}(Q)$ *solving*

$$(\partial_t - a\partial_{yy} - b\partial_y + \lambda)v = g.$$

If a past-zero approximation $\widetilde{v}$ *in the same space has residual* $r = g - (\partial_t - a\partial_{yy} - b\partial_y + \lambda)\widetilde{v}$, *then*

$$\|\partial_{yy}(v-\widetilde{v})\|_p \le \frac{A_p}{1-\kappa}\|r\|_p, \quad \|\partial_y(v-\widetilde{v})\|_p \le \frac{\lambda^{-1/2}}{1-\kappa}\|r\|_p. \tag{2.4}$$

For $p > 3$, *also*

$$\|\partial_y(v-\widetilde{v})\|_\infty \le \frac{K_p\lambda^{-(p-3)/(2p)}}{1-\kappa}\|r\|_p, \quad K_p = \|1_{t>0}e^{-t}\partial_y G_t\|_{L^{p'}(\mathbb{R}^2)} < \infty. \tag{2.5}$$

Proof. Write $v = T_\lambda f$ and $\mathcal{B}f = (a-1)\partial_{yy}T_\lambda f + b\partial_y T_\lambda f$. Equations (2.2) give $\|\mathcal{B}\| \le \kappa$. Thus $f = (I-\mathcal{B})^{-1}g$ exists by the norm-convergent Neumann series, and $\|f\|_p \le \|g\|_p/(1-\kappa)$. Maximal regularity and $v_t = f + v_{yy} - \lambda v$ give the named space and zero initial trace. Conversely every past-zero solution in that space is the causal heat potential of its constant-coefficient forcing, so the same contraction proves uniqueness. Applying the argument to the error gives (2.4).

For (2.5), Holder's inequality bounds the causal convolution from L^p to L^∞. At fixed time, $\int |\partial_y G_t|^{p'}\,\mathrm{d}y = C_{p'}t^{1/2-p'}$. The integral at zero is finite exactly when $p' < 3/2$, or $p > 3$. Scaling $y \mapsto \sqrt{\lambda}y, t \mapsto \lambda t$ gives the stated power of λ. The infinite spatial complement is part of this kernel norm, not discarded. □

The proof is classical perturbative inversion, credited as such. Its use here is to name the precise causal residual and constants that a finite certificate would have to bound. The J-term forcing iterate $f_J = \sum_{j=0}^{J}\mathcal{B}^j g$ has remaining norm at most $\kappa^{J+1}\|g\|_p/(1-\kappa)$; this is an infinite-function-space bound, not a finite-rank theorem. Any numerical approximation to the individual terms has its own residual.

For an undamped equation one may conjugate by $e^{-\lambda(t-t_0)}$. Restoring the original solution costs at most $e^{\lambda(t_1-t_0)}$, so an unlimited time interval is not covered by a uniform claim. Likewise a cutoff uses (2.1); the term involving H_y on its collar cannot be deleted merely because the interior estimate is derivative-free in the coefficients. There is no imposed terminal-zero trace. Imposing one would generally overdetermine the causal problem.

3 Moving cylinders and the physical reaction matrix

The source uses $h_c = r_c/\sqrt{q}$,

$$t = t_c + h_c^2\rho, \quad r = R_c(t) + h_c y, \quad R_c(t) = \sqrt{r_c^2 + 2(q-1)(t_c - t)}.$$

For $q \ge 16$, $r_c \le \sqrt{T-t_c}/8$, and $(y,\rho) \in P_2$, one has $r_c/2 \le r \le 7r_c/2$ and $r^2 < \delta(t)$. The rescaled drift for $H = q(\sqrt{z}-1)$ is

$$b = h_c(q-1)(1/r - 1/R_c) - h_c z/r.$$

Its first term equals $-(q-1)h_c^2 y/(rR_c)$, which is uniformly bounded. Without movement of the center the corresponding term is of order $\sqrt{q}$. The source term uses $qh_c^2 = r_c^2$, $(q-1)|1-z| \le C_0$, and $r|u_r| \le 2$. This explains why the one-dimensional heat constants are independent of the ambient dimension. A second, smaller cylinder applied to $r_c u_r$ uses the first derivative estimate

before differentiating the u equation. This is not a simultaneous assumption of both derivative bounds.

Under the source curvature convention, $\langle \mathrm{Rm}(D), D\rangle = \sum_{i,j} R_{ijij} D_i D_j$. In physical coordinates $(\alpha, \sqrt{2}\beta, \sqrt{q}\chi)$, the curvature operator on invariant diagonal tensors is

$$\mathcal{R} = \begin{pmatrix} 0 & \sqrt{2}h & \sqrt{q}h_q \\ \sqrt{2}h & j & \sqrt{2q}\mu \\ \sqrt{q}h_q & \sqrt{2q}\mu & (q-1)\ell \end{pmatrix}. \tag{3.1}$$

For example the cross terms are $4h\alpha\beta$, $2qh_q\alpha\chi$, and $4q\mu\beta\chi$, exactly the ordered sectional-curvature contractions. With the evolving metric included,

$$(\partial_t - \Delta)|\operatorname{Ric}|^2 = -2|\nabla \operatorname{Ric}|^2 + 4\langle \mathrm{Rm}(\mathrm{Ric}), \mathrm{Ric}\rangle.$$

Consequently, at nonzero Ricci norm, Kato's inequality gives

$$(\partial_t - \Delta)|\operatorname{Ric}| \le c|\operatorname{Ric}|, \qquad c = \frac{2\langle \mathrm{Rm}(\mathrm{Ric}), \mathrm{Ric}\rangle}{|\operatorname{Ric}|^2} \le 2\lambda_{\max}(\mathcal{R}). \tag{3.2}$$

This upper bound is valid for the actual Ricci vector; it does not identify that vector with a maximizing eigenvector.

Lemma 3.1 (Separated-block envelope)**.** *Let a real symmetric matrix, in an orthogonal splitting, be* $A = \left(\begin{smallmatrix} \alpha & b^T \\ b & D \end{smallmatrix}\right)$*. If* $\alpha \le a$*,* $D \preceq dI$*, and* $\|b\| \le v$*, then*

$$\lambda_{\max}(A) \le \Phi(a, d, v) := \frac{a + d + \sqrt{(a-d)^2 + 4v^2}}{2}. \tag{3.3}$$

If $a > d$*, this is at most* $a + v^2/(a-d)$*. The first bound is optimal using only these three hypotheses.*

Proof. For a vector (x, y), its quadratic form is at most $a|x|^2 + 2v|x|\|y\| + d\|y\|^2$. The largest Rayleigh quotient of this two-dimensional form is (3.3). For $g = a - d > 0$, the excess over a equals $2v^2/(\sqrt{g^2 + 4v^2} + g) \le v^2/g$. Taking $D = dI$ and coupling one unit direction by v gives equality in the first bound. This is classical Rayleigh/Schur algebra; the source-specific issue is which block has a dimension-sized gap. □

Theorem 3.2 (Constant-order reaction loss)**.** *Assume* (1.3)–(1.4)*, with* q *large enough that* $z \le 2$*. Define dimension-independent constants*

$$H_0 = 2U + 8 + Z, \quad J = Z/2, \quad M_0 = 4, \quad B_0^2 = 2(H_0^2 + M_0^2), \quad d_0 = C_0 + J, \quad a_q = q - 1 + C_0. \tag{3.4}$$

On the same region as (1.4)*,*

$$cr^2 \le 2\Phi(a_q, d_0, B_0\sqrt{q}). \tag{3.5}$$

If in addition $q \ge 2(J+1)$*, then*

$$c \le \frac{2(q-1) + C_R}{r^2}, \qquad C_R = 2C_0 + 4B_0^2. \tag{3.6}$$

All constants are independent of q, k*, although their numerical values require quantitative versions of the inherited heat estimates.*

Proof. The formulas (1.2) imply

$$|r^2 h| \le 2U\sqrt{q} + 8 + Z/\sqrt{q} \le H_0\sqrt{q}, \quad |r^2 h_q| \le J/\sqrt{q}, \quad |r^2\mu| \le M_0.$$

Also $r^2 j \le q - 1 + C_0$ and $(q-1)r^2\ell \le C_0$; no lower bounds for these diagonal entries are needed. Reorder $r^2\mathcal{R}$ so that its β coordinate comes first. The remaining block is

$$D = \begin{pmatrix} 0 & \sqrt{q}\, r^2 h_q \\ \sqrt{q}\, r^2 h_q & (q-1)(1-z) \end{pmatrix} \preceq (C_0 + J)I.$$

The coupling vector has squared norm $2r^4h^2 + 2qr^4\mu^2 \le B_0^2 q$. Apply Lemma 3.1 and (3.2). Finally $a_q - d_0 = q - 1 - J \ge q/2$, so the spectral excess is at most $2B_0^2$, proving (3.6). □

Source `05-reaction.tex`, lines 250–256, bounds the entire off-diagonal part by its $O(\sqrt{q})$ norm. Retaining the gap between the dominant β diagonal and the other two coordinates replaces that loss by $O(1)$. The matrix, weighting, derivative estimates and factor two in the Ricci evolution are unchanged. The increment is not the classical block lemma in isolation.

Example 3.3 (Why a constant loss remains)**.** The matrix $\left(\begin{smallmatrix} q & \sqrt{q} \\ \sqrt{q} & 0 \end{smallmatrix}\right)$ has excess above q

$$\frac{\sqrt{q^2 + 4q} - q}{2} \longrightarrow 1.$$

For $q \ge 1$, this excess lies between $1 - 1/q$ and 1: substitute those two numbers into $s(q+s) - q$, which is increasing for $s \ge 0$. Thus the stated size hypotheses alone cannot produce an $o(1)$ loss. The fixture is a symmetric matrix, not an asserted realization by a doubly warped Ricci flow or its actual Ricci vector.

Proposition 3.4 (Block-aware uncertainty)**.** *Suppose a computed physical block has certified upper bounds $\widetilde{\alpha} \le a$, $\widetilde{D} \preceq dI$, $\|\widetilde{b}\| \le v$. If the true block errors satisfy $\alpha - \widetilde{\alpha} \le \epsilon_a$, $D - \widetilde{D} \preceq \epsilon_d I$, and $\|b - \widetilde{b}\| \le \epsilon_b$, then*

$$\lambda_{\max}(A) \le \Phi(a + \epsilon_a, d + \epsilon_d, v + \epsilon_b).$$

Proof. The three hypotheses of Lemma 3.1 hold with the enlarged bounds. No claim is made that sampled values supply uniform errors. □

This formula distinguishes weighted tensor errors from coordinate errors. In particular the mixed entry includes $\sqrt{2q}\mu$. Ignoring that weight is not justified by small raw coefficient residuals.

4 The two-scale barrier with the new budget

For a scalar composition in the radius, the source's geometric operator is

$$\mathcal{D}F = (\partial_t - \Delta)F = F_t|_r - zF_{rr} - (q - 1 + z)F_r/r. \tag{4.1}$$

Although the radius-coordinate PDE is only used in a hemisphere, this composition identity extends across the equator. It follows from $r_t - \Delta r = -(q - 1 + z)/r$, with the time derivative on the left taken at a fixed manifold point.

Theorem 4.1 (Conditional refined Ricci comparison)**.** *Assume one smooth source-type flow has $z \le 2$ on its inner region, the norm inequality (3.2),*

$$c \le [2(q-1) + C_R]/r^2, \qquad c \le C_*/\theta^2,$$

and the stated parabolic-annulus Ricci estimates. Let $a > d_q$, $L = k - 1 > 0$, $\sigma = k/d_q$, and suppose

$$0 < e := a - d_q + d_q/k < 1, \quad \nu > 1, \quad \mathfrak{m} = (a-2)(q-1) - C_R - 2a^2 - 1 > 0. \tag{4.2}$$

Choose $\xi \le \min\{1/8, 1/(2\sqrt{L})\}$*, and, after the flow and* C_* *are fixed, choose* $b_0 > 0$ *with* $b_0 C_* < \mathfrak{m}$*. At all sufficiently late times,*

$$|\operatorname{Ric}| \le C\delta^L (r^2 + b_0\theta^2)^{-a/2} \quad (0 \le r \le \xi\sqrt{\delta}), \qquad |\operatorname{Ric}| \le C' r^{-e} \quad (r > 0 \text{ in this region}). \tag{4.3}$$

The constants in the conclusion are for this flow and are not uniform over the construction parameters.

Proof. Put $D_0 = r^2 + b_0\theta^2$ and $H = \delta^L D_0^{-a/2}$. It is smooth at the pole because it is a smooth function of r^2. Direct differentiation in (4.1) gives

$$\frac{\mathcal{D}H}{H} = -\frac{L}{\delta} + \frac{a\sigma b_0\theta^2}{\delta D_0} + \frac{a(q-1) + 2az}{D_0} - \frac{a(a+2)zr^2}{D_0^2} \tag{4.4}$$

$$\ge -\frac{L}{\delta} + \frac{a(q-1) - a^2 z}{D_0}. \tag{4.5}$$

Since $\sigma > 1/2$, at sufficiently late times $LD_0/\delta \le L\xi^2 + Lb_0\delta^{2\sigma-1} \le 1$. For $r > 0$, add the two valid reaction upper bounds after multiplying by r^2 and $b_0\theta^2$, respectively. Thus

$$cD_0 \le 2(q-1) + C_R + b_0 C_*.$$

This remains an upper bound even when $c < 0$. At a pole the cap bound alone suffices. Hence

$$\mathcal{D}H/H - c \ge (\mathfrak{m} - b_0 C_*)/D_0 > 0.$$

On $r = \xi\sqrt{\delta}$, $H \asymp \delta^{L-a/2}$, so $|\operatorname{Ric}|/H \le C\delta^{\nu-1-L+a/2} = C\delta^{(a-d_q)/2}$ is bounded. One multiplier dominates lateral data and a fixed late starting slice. At a first contact with this positive barrier, $|\operatorname{Ric}| > 0$ and its norm is smooth; the strict inequality excludes contact. The poles are interior manifold points, not additional boundary data. Finally $\delta^L = \theta^{a-e}$. Splitting at $r = \theta$ gives $H \le r^{-e}$ outside and $H \le b_0^{-a/2}\theta^{-e} \le b_0^{-a/2} r^{-e}$ inside. □

Corollary 4.2 (A dimension-sensitive exponent schedule)**.** *Let* $\kappa = C_R + 20$*, and choose an integer* q *satisfying all source universal thresholds,* $q \ge 12$*, and* $q - 1 \ge 2\kappa$*. Set*

$$a = d_q + \kappa/(q-1).$$

For every sufficiently large even k*, the preceding theorem applies with* $e = \kappa/(q-1) + d_q/k$*, once its geometric inputs are supplied.*

Proof. The number d_q is the smaller root of $d^2 - (q+1)d + 2(q+1) = 0$. For $q \ge 12$ it lies in $(2, 5/2)$: the polynomial is positive at two and negative at $5/2$, and its vertex lies to the right. Thus $a < 3$, $a > d_q$, and

$$\mathfrak{m} = (d_q - 2)(q-1) + \kappa - C_R - 2a^2 - 1 > 1.$$

For large even k, $\nu > 1$, $\sigma > 1/2$, and $0 < e < 1$; the first summand of e is at most $1/2$. Choose the construction only afterward, followed by b_0. □

For comparison, the source fixes $a = 5/2$, giving $e = 5/2 - d_q + d_q/k$; its displayed reaction margin contains $C\sqrt{q}$. Reoptimizing that coarser margin would naturally cost $a - 2$ of order $q^{-1/2}$. The retained-block estimate permits an order q^{-1} loss. It does not provide a numerical minimal dimension, since U, Z, and the source universal thresholds have not been numerically certified. Nor does it assert a single flow with all these dimensions or uniform constants as $q \to \infty$.

The cap bookkeeping above cannot be reduced merely by writing a minimum: for $A, C, b > 0$,

$$\sup_{x>0} \min\{A(1 + b/x), C(x + b)\} = A + bC,$$

with equality at $x = A/C$. Thus adding the two cap/radius bounds is already optimal if those are the only available inputs.

4.1 Why scalar comparison still needs the same pole argument

If $|\operatorname{Ric}| \le K_R r^{-e}$, put $\ell_0 = 2 - 2e \in (0,2)$. Equation (4.1) gives

$$\mathcal{D}(-r^{\ell_0}) = \ell_0(q - 1 + \ell_0 z)r^{-2e}, \qquad \mathcal{D}(r^{-1}) = (q - 1 - z)r^{-3}.$$

Consequently $C_1 - C_2 r^{\ell_0} + \epsilon/r$ dominates the scalar source $2|\operatorname{Ric}|^2$ when $C_2 \ge 2K_R^2/[\ell_0(q-1)]$ and C_1 controls initial/lateral data. The power r^{ℓ_0} need not be smooth at the pole. For fixed $\epsilon > 0$ and a fixed compact slab ending at $t_1 < T$, comparison first takes place on an excised domain and its inner radius tends to zero. Then $\epsilon \downarrow 0$ pointwise on that slab; C_1 is independent of both ϵ and t_1. The arbitrariness of $t_1 < T$ gives the bound for every $t < T$, and continuity at each such time supplies the pole value. No fractional barrier is evaluated at T. Our sharper reaction bound does not remove these ordered-limit or geometric prerequisites.

5 Worked certificates and necessary distinctions

Example 5.1 (A genuinely coupled finite reaction fixture)**.** Take $q = 16$, $a = 16$, $d = 1$, and coupling norm $v = 12$. The two-by-two matrix $\left(\begin{smallmatrix} 16 & 12 \\ 12 & 1 \end{smallmatrix}\right)$ has exact top eigenvalue $(17 + \sqrt{801})/2$. Lemma 3.1 gives this exact envelope and the rational upper bound

$$16 + \frac{144}{15} = \frac{128}{5}.$$

For the latter certificate the diagonal entries of $(128/5)I - A$ are $48/5, 123/5$, and its determinant is $2304/25 > 0$. Thus this upper bound is verified without a numerical eigenvalue routine. It is a fixture for the algebra, not a sampled curvature tensor from the constructed flow.

Example 5.2 (A causal error budget)**.** Suppose $A_p\varepsilon \le 1/10$, $B = 2$, and $\lambda = 100$. Then $\kappa \le 3/10$. A residual of L^p norm at most $1/100$ gives gradient error at most $1/700$ by (2.4). The forcing-series tail after $J = 3$ is at most $81/7000$ times the original forcing norm. These are exact conditional constants. They do not certify the value of A_p, the size of a geometric coefficient error, or a computed quadrature.

Example 5.3 (Energy control alone is insufficient)**.** Take nonzero smooth compactly supported χ, ψ, put $\varphi = \chi'$, and choose a time t_* strictly inside a fixed cylinder. For large integers n, define

$$H_n(y,t) = \varphi(ny)\psi(n^2(t - t_*)).$$

Its spatial gradient supremum grows as n, while

$$\|H_n\|_{L_t^\infty L_y^2} + \|\partial_y H_n\|_{L_{y,t}^2} + \|\partial_t H_n\|_{L_t^2 H_y^{-1}} = O(n^{-1/2}).$$

The first two statements follow by changing variables. For the third, $\partial_t H_n = n\partial_y[\chi(ny)]\psi'(n^2(t - t_*))$; the H^{-1} norm of a derivative is at most the L^2 norm of its primitive. Time integration contributes another factor n^{-1}. One can choose the supports inside a fixed bounded spatial domain, so the same estimate holds for the dual of H_0^1. These smooth functions are not a counterexample to the source heat lemma: their pointwise forcing is not uniformly bounded. They show why its stronger gradient conclusion cannot be read off from energy-space control alone.

A second elementary negative control concerns physical weights. With only mixed curvature $\mu = 1$, $q = 8$, $\beta = 2$, $\chi = 1$, the actual curvature quotient on the two fiber components is $4q\beta\chi/(2\beta^2 + q\chi^2) = 4$. Treating the raw coefficients as a Euclidean two-by-two matrix with off-diagonal one instead gives $2\beta\chi/(\beta^2 + \chi^2) = 4/5$. This is not a harmless change of notation; the corresponding symmetric operator must be conjugated using the physical Gram matrix $\operatorname{diag}(1, 2, q)$.

6 Original Ram proofs, scope, and practical consequences

The freshly read original TR8 [6], lines 81–177, proves a coercive inverse on H_0^1 with load space H^{-1}, states the weak operator domain, and retains the boundary lifting in its Green equivalence. Lines 178–236 give an explicit variable-coefficient one-dimensional kernel, the physical load/mass distinction for finite systems, and the spectral heat/Duhamel equivalence in energy spaces. We read through line 240, but do not use the beginning of the following wave section as a completed proof. These are original local proof bodies, not catalogue locators.

The connection here is causal inversion with the actual operator and error source retained. The whole-line kernel in Theorem 2.1 is not TR8's bounded-domain Dirichlet kernel; neither its derivative L^p bound nor parabolic embedding follows from the energy theorem. The extra maximal-regularity estimate is explicitly external. Once it is given, the coefficient multiplication and Neumann remainder are proved on the full causal space. No finite jet representation eliminates the spatial tail, initial trace, or cutoff collar.

The freshly read secondary application TR-item-047 [7], lines 346–412, shows how a genuine polynomial identity defect and a Gram-positivity defect must be charged separately on a proved domain. It motivates the reporting discipline in Proposition 3.4, but supplies neither a Ricci-flow theorem nor a parabolic estimate. Its finite optimization examples are not counted as our checks.

6.1 Advantages and costs

The primary improvement keeps exactly the source's physical reaction operator. It uses a dimension-sized spectral gap rather than discarding the block structure, reduces the algebraic loss to constant order, and permits a correspondingly finer barrier exponent schedule. It does not require differentiating additional geometric coefficients or changing the cap construction. The block-aware error form also allows independently certified uncertainties to be carried through without declaring an approximate identity exact.

There are important costs. The universal heat constants are not numerically supplied, so even the sharper formula does not name the lowest admissible dimension. The cap constant C_*, starting time, barrier multipliers and profile errors still belong to an entire chosen flow. A causal residual certificate requires a norm over the full space-time domain; a finite collection of point residuals is insufficient. Taking a large damping parameter improves contraction but can make the undamped finite-time estimate poor. There is no measured speedup claim.

6.2 Precisely what remains imported

Stolarski's existence construction and the translation of its profile classes are not reconstructed here. The source's global ratio and point-picking arguments, controlled parabolic neighborhoods for Shi's estimates, pole-orbit lower size, cap slope and all required late-time thresholds remain geometric obligations. The heat maximal-regularity theorem is a distinct analytic obligation. Our local reaction theorem is conditional on (1.3)–(1.4); the comparison theorem additionally names its cap and boundary inputs.

In particular, a verified three-by-three inequality cannot prove the global singularity claim. Bounded scalar curvature is not deduced from a trace cancellation in one tangent space. Smoothness on every compact time slab is not uniform regularity at T. Nor may one let q, k, or the cap parameter vary with time inside the already chosen flow. Our contribution is a strengthened conditional estimate in the existing proof architecture, not a second construction of the example.

Future work could certify explicit one-dimensional heat constants and track the source's universal thresholds numerically, or use a sharper actual Ricci-vector bound instead of the top eigenvalue over all invariant tensors. Either route requires additional rigorous input. For a variable reference diffusion, a new causal inverse estimate and its coefficient norm would have to

be established before the same perturbation argument applies. These are possible extensions, not completed results.

7 Current primary literature and evidence

The public status checks in this report were made on October 9, 2026. Stolarski's arXiv record [2] is v1, April 30, 2019; its abstract states high-dimensional doubly warped conical Type-II formation, not the scalar bound asserted by the assigned manuscript. The assigned source uses numbered references to that version and identifies the later Memoirs publication. Its full external proof is imported here, not independently audited.

Buzano–Di Matteo [4], v1 of June 10, 2026, state in their abstract that a Type-I singular point forces Type-I scalar blowup, without a global Type-I hypothesis. This is consistent with the source's Type-II target, not confirmation of its construction. Liu–Simon [5], published April 13, 2026, study spatial L^p scalar control with $p > n/2$; their abstract distinguishes four-dimensional orbifold continuation from smooth continuation on the original closed manifold. We checked these primary abstracts and version/publisher records, not their full proofs.

For Krylov [3], the primary v1 abstract and the statement of Theorem 5.5 were read, including its whole-space derivative estimate, nonnegative spectral parameter and time convention. Its full proof and the source's cited Sobolev embedding remain external analytic inputs. No confidential corpus text or title was sent to public search services.

The read ledger records 11 assigned public text files, totaling 1,549 literal lines, with no unread complement in that inventory. These include all six scientific modules. Fresh private scope is TR8 81–240 and TR-item-047 346–412, totaling 227 lines; the exact inventoried unread complement totals 1,420 lines. Other private files, the external construction's full proof, and other repository manuscripts are outside the fresh read claim. PRE hashes were persisted before the first scientific display; POST hashes verify that the sources have not changed. Heading searches and truncated discovery output are not counted as literal proof reading.

The owned executable `M676-checks.py` passed 650 exact rational cases and 2,780 counted assertions, with no floating-point diagnostic cases. It checks ordered curvature contractions, physical norms, nontrivial three-by-three Schur certificates, moving-cylinder identities, automatic differentiated two-scale barriers, scalar powers, exponent budgets, cap minima and causal geometric tails. The exact derivative jets in the test are finite algebra, not a Ricci-flow integrator. The constant-loss and unweighted-coordinate fixtures are negative controls. The script SHA and result are linked in the owned receipt. No parent, earlier-report or foreign checker cases are included.

8 Conclusion

The useful retained information is the large spectral separation already present in the source reaction matrix. It changes the local dimension loss from $O(\sqrt{q})$ to $O(1)$ and supports an explicit q^{-1}-scale barrier budget, with all geometric inputs still named. Causal residual estimates clarify how a quantitative operator certificate would have to control the continuum complement. These demonstrated local deductions justify C3 independently of L3 applicability; they do not independently settle the scalar-curvature extension question.

References

[1] OpenAI, *A closed Ricci flow with bounded scalar curvature and finite-time curvature blowup*, September 24, 2026. Pinned public manuscript. Assigned scientific modules read in full; global claim attributed.

[2] M. Stolarski, *Curvature Blow-up in Doubly-warped Product Metrics Evolving by Ricci Flow*, arXiv:1905.00087v1, 2019. Primary abstract/version record checked. External construction proof not independently audited here.

[3] N. V. Krylov, *Parabolic equations with VMO coefficients in spaces with mixed norms*, arXiv:math/0610955v1, 2006, Theorem 5.5. Published in J. Funct. Anal. 250 (2007), 521–558. Primary statement and convention inspected; full proof imported.

[4] R. Buzano and G. Di Matteo, *A Local Singularity Analysis for the Ricci Flow and its Applications to Ricci Flows with Bounded Scalar Curvature—Part II*, arXiv:2606.12379v1, June 10, 2026. Abstract and version record checked.

[5] J. Liu and M. Simon, *Volume estimates and convergence results for solutions to Ricci flow with L^p bounded scalar curvature*, Calc. Var. PDE 65 (2026), article 161. Publisher abstract and metadata, April 13, 2026.

[6] M. Subbarao, confidential original TR8, local directory `RamNonlin-IE-PDE-astra/TR8-lin-non-lin`, file `TR8-lin-non-lin.tex`; fresh lines 81–240. Original weak inverse, physical Green and energy heat proof bodies.

[7] M. Subbarao, confidential secondary application, file `TR-item-047-astra.tex`, local `res-rt/res-math`; fresh lines 346–412. Residual-aware Gram certificate comparison, not an original parabolic regularity theorem.